

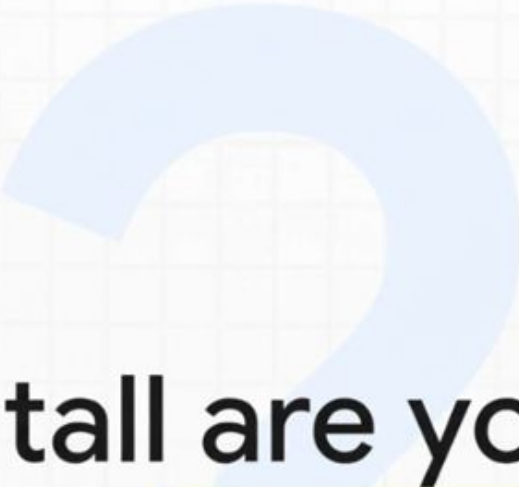
The Tale of Two Rulers

IMPERIAL



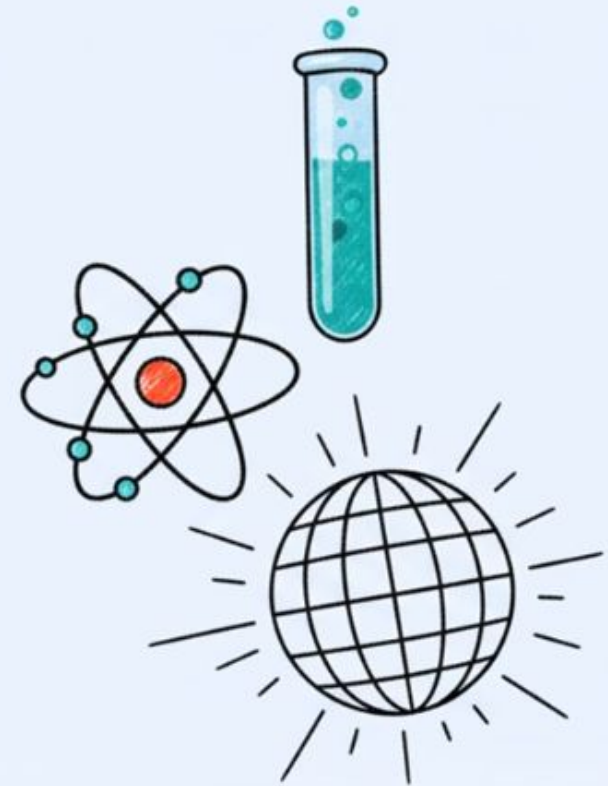
METRIC





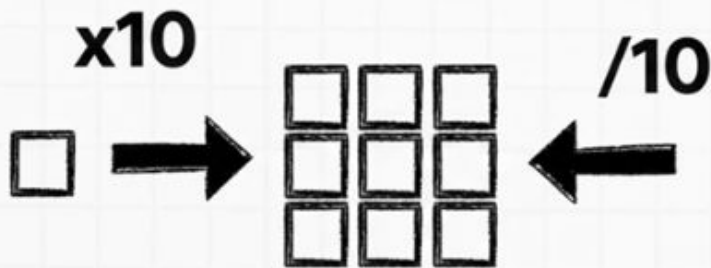
How tall are you? In
feet or in **centimeters**?







10





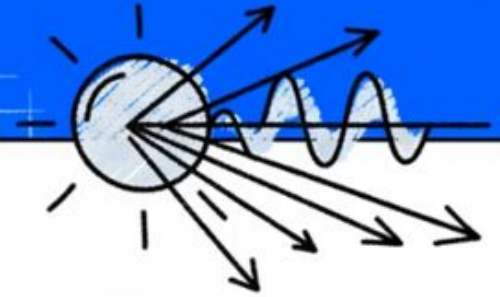
Prefix	Meaning	Relation to Meter
Kilo- (k)	1000	10^3
Centi- (c)	1/100	10^{-2}
Milli- (m)	1/1000	10^{-3}





The Meter's Journey

A Universal Ruler



Evolution of the Meter



1790s

Defined as 1/10,000,000 of the distance from the North Pole to the Equator.

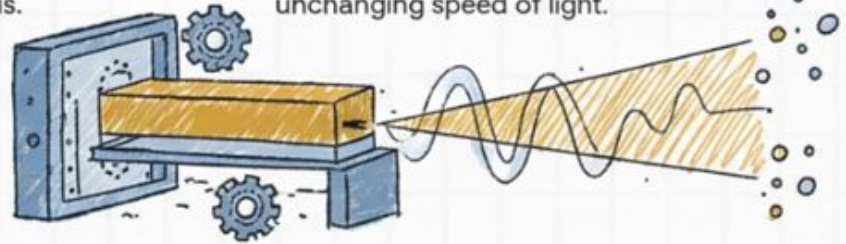


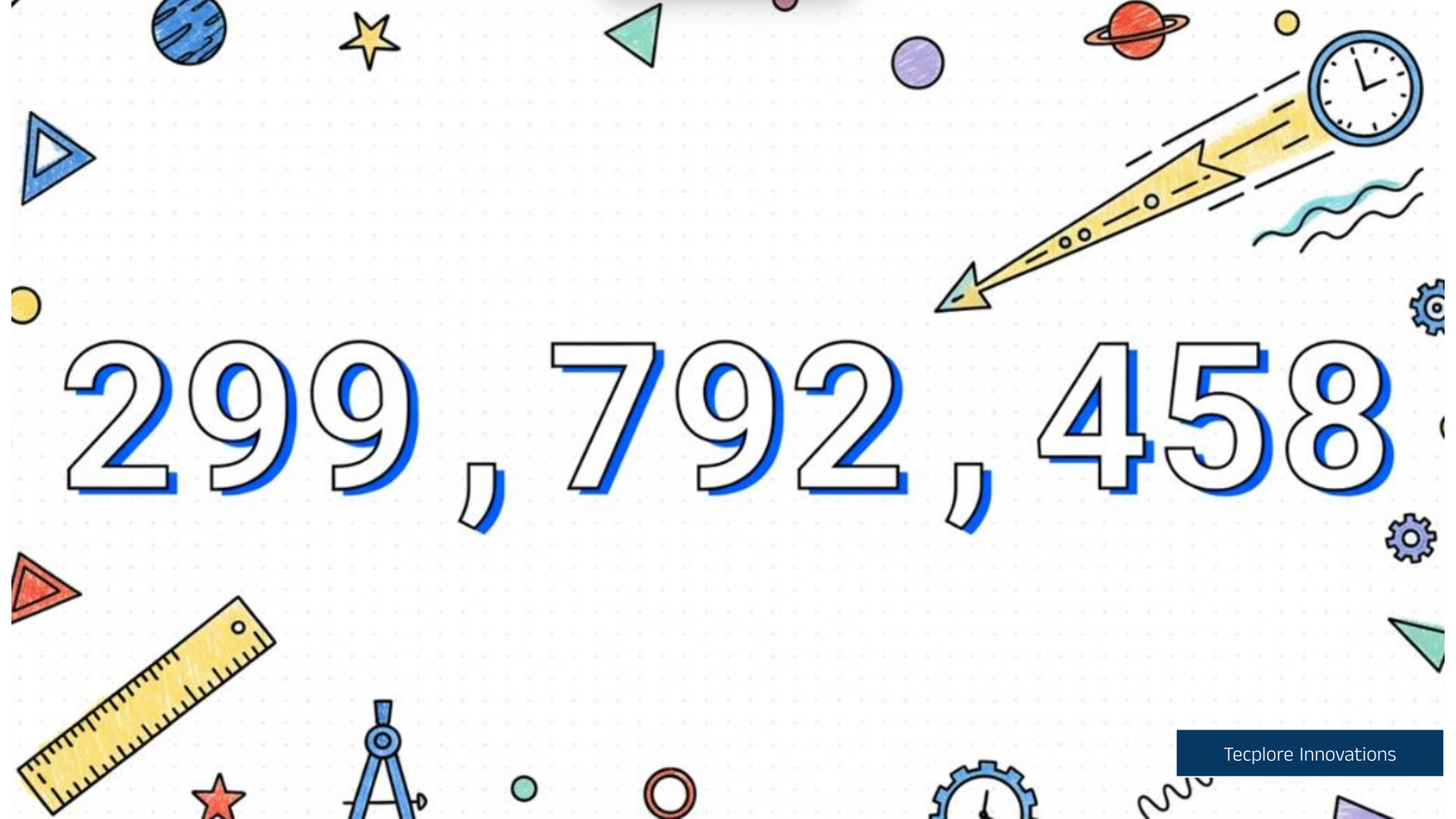
1889

The physical length of a Platinum-Iridium bar kept in a vault in Paris.

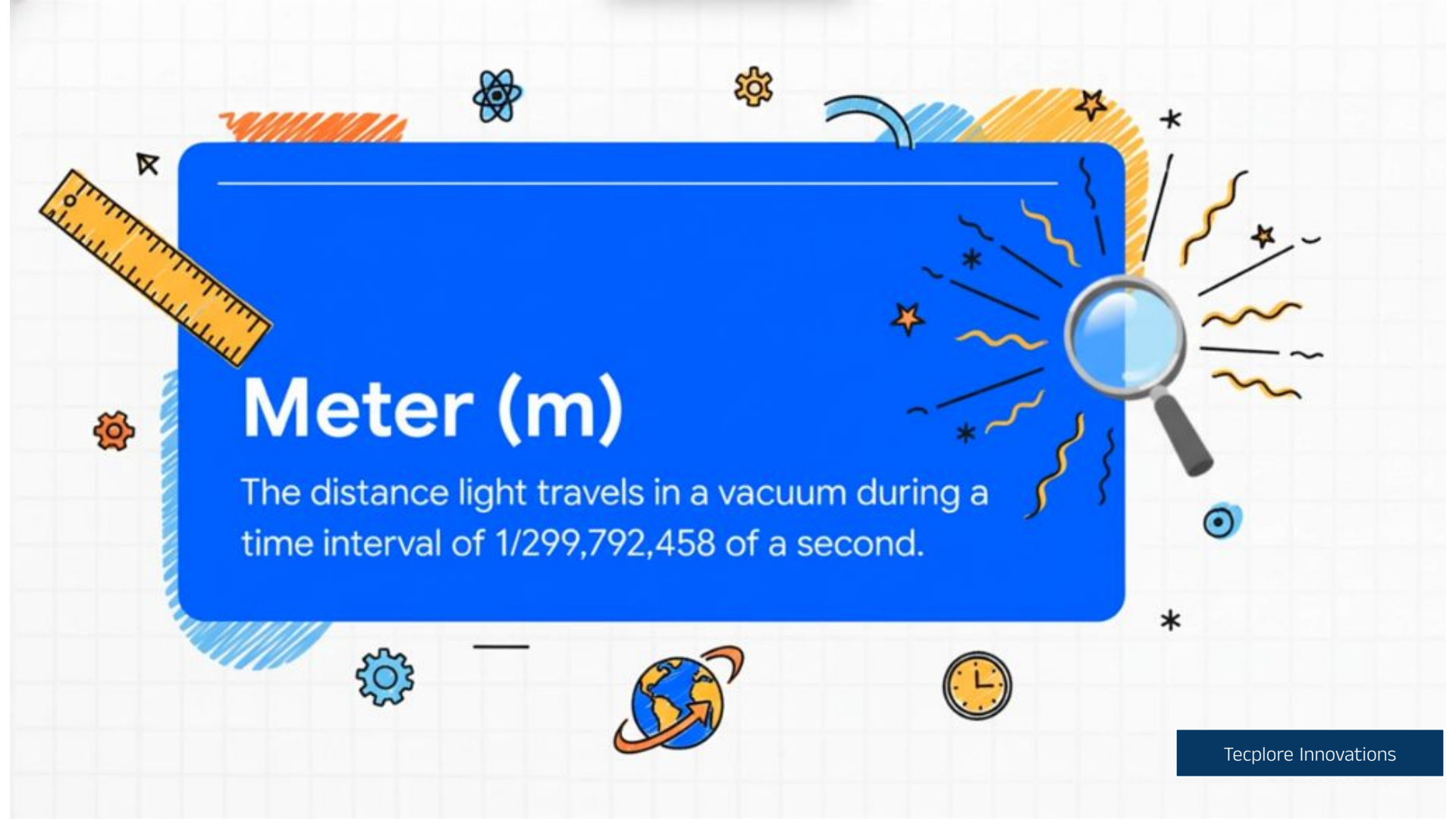
1983

Defined by a universal constant: the unchanging speed of light.





299,792,458



Meter (m)

The distance light travels in a vacuum during a time interval of $1/299,792,458$ of a second.

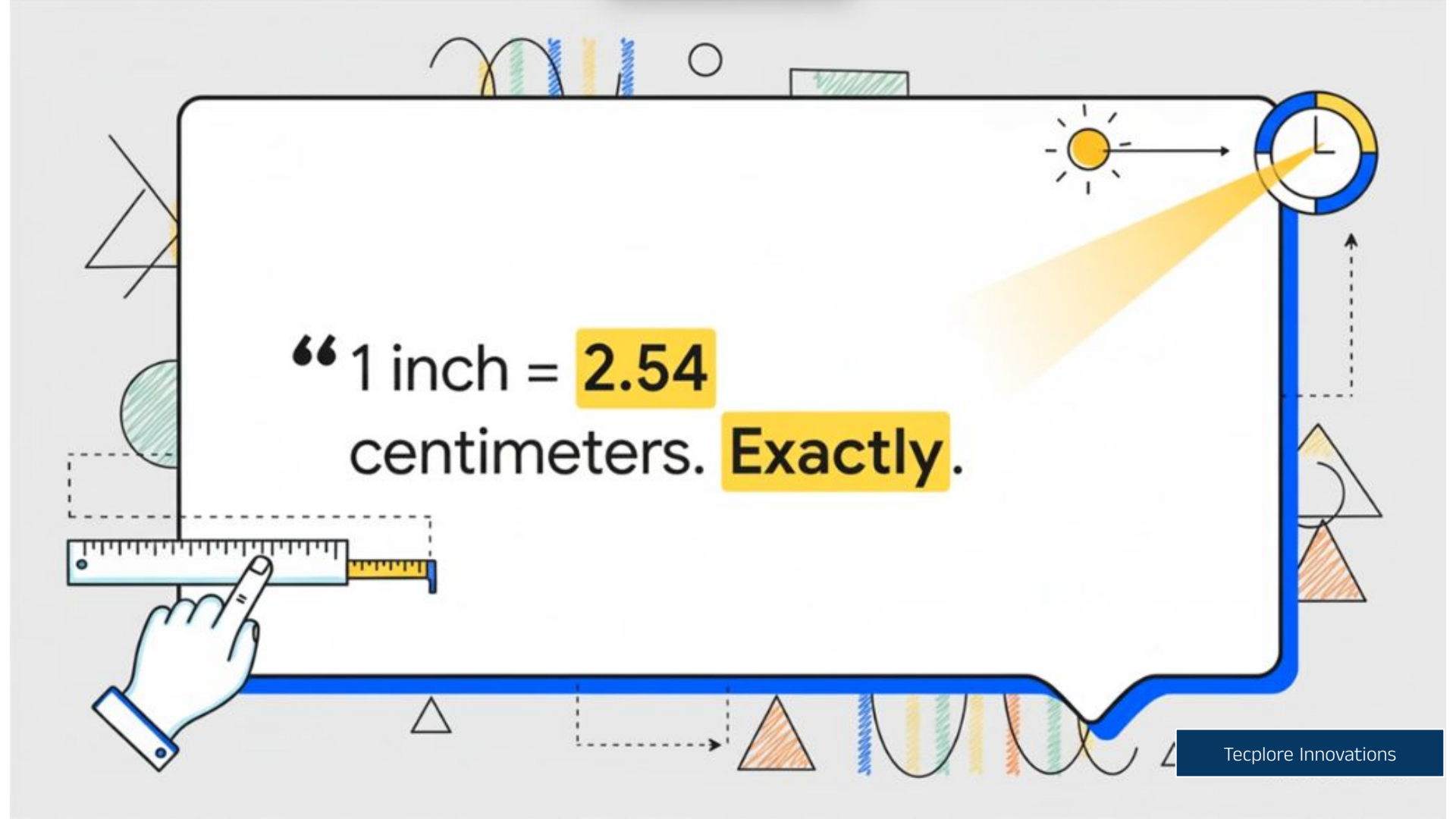
2



The Conversion Bridge

Connecting Two Worlds





“1 inch = 2.54
centimeters. Exactly.

Feet to Meters

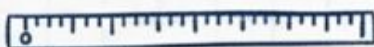


1. Feet to Inches

Multiply by 12 (e.g., 5 ft * 12 = 60 in)



12x



2. Inches to Cm

Multiply by 2.54 (e.g., 60 in * 2.54 = 152.4 cm)



2.54x

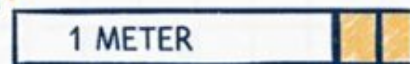
/100

3. Cm to Meters

Divide by 100 (e.g., 152.4 cm / 100 = 1.524 m)



af





3

Why It Matters

The Takeaway

